Ameer Sohail Shaik

AI Engineer | Forward Deployed Engineer | LLM Agents, RAG, Microsoft Fabric, Python

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PROFESSIONAL SUMMARY

M.S. Data Science student who shipped three production LLM systems as an AI Intern at Oldcastle BuildingEnvelope: a Microsoft Fabric data agent for customer service (97%+ accuracy) that replaced an estimated $250K+/year Snowflake Cortex build, a source-cited RAG chatbot over 4,000+ documents (95%+ on 600+ questions), and an Agent Skill automating Accounts Payable invoice coding. Strong in Python, SQL, RAG, agent design, and LLM evaluation.

KEY SKILLS

LLM Agents & Applications: Microsoft Fabric Data Agent, Microsoft Copilot Studio, Claude Agent Skills, LangChain, RAG pipelines, Text-to-SQL, prompt engineering, tool use, human-in-the-loop workflows, LLM evaluation (LLM-as-judge, test sets), source-cited answers

LLM APIs & Retrieval: OpenAI/GPT, Anthropic Claude, Gemini, Groq APIs; ChromaDB, sentence-transformers, hybrid retrieval (dense + BM25), Reciprocal Rank Fusion, cross-encoder re-ranking, OCR document ingestion, metadata crosswalks

Programming & Querying: Python (pandas, NumPy, scikit-learn), SQL (joins, window functions, CTEs), Java, Bash; Git, Agile development; PostgreSQL, MySQL, SQLite

Data Platforms & Cloud: Microsoft Fabric, OneLake, Snowflake, Microsoft Teams integration; ETL pipelines, PySpark, Apache Airflow, schema validation, data quality checks; AWS, GCP, Azure (foundational)

Machine Learning & Analytics: PyTorch, TensorFlow, XGBoost, Random Forest, Gradient Boosting; classification, forecasting, feature engineering, hyperparameter tuning, SHAP, A/B testing; Power BI (DAX, Power Query), Streamlit

PROFESSIONAL EXPERIENCE

AI Intern | Oldcastle BuildingEnvelope (OBE) | Dallas, TX *Jun 2026 – Aug 2026*

- Built the CRL 360 agent as a Microsoft Fabric Data Agent in Microsoft Teams, letting customer service reps ask backlog, order, and delivery questions in plain English over 9 tables of 5M+ rows each; bridged Snowflake to OneLake so the agent ran on existing licensing, replacing a proposed Snowflake Cortex Agent (estimated $250K+/year for 300 users) at no additional licensing cost.
- Validated the agent on 170+ real customer service questions through manual review by senior data engineers and LLM-based evaluation, reaching 97%+ answer accuracy before rollout and removing manual report pulls for complex, multi-table questions.
- Designed and deployed a Retrieval-Augmented Generation (RAG) chatbot on Microsoft Copilot Studio in Microsoft Teams over 4,000+ technical documents and 188K+ product SKUs; built the ingestion pipeline (OCR, SKU-to-document metadata crosswalk, semantic retrieval) so every answer cites its source, scoring 95%+ accuracy on 600+ evaluation questions and cutting lookup time from minutes to seconds.
- Automated weekly invoice coding and use tax self-assessment for the Accounts Payable team by encoding vendor-specific allocation rules into an Agent Skill deployed on Claude and Microsoft Copilot; analysts attach invoices and get an upload-ready Excel workbook with per-vendor sheets for review, live for 15+ weekly and 5 monthly vendors, saving about 6 hours/week plus 5+ hours per month-end cycle.

Junior Data Scientist | Proceedit | Remote *Aug 2024 – Aug 2025*

- Designed an offline evaluation framework in Python comparing 10 ML architectures on directional accuracy, Sharpe ratio, and max drawdown; gradient-boosted models matched deep learning precision (65%) at 12x lower inference cost, blocking an over-engineered LSTM from production.
- Ran feature importance analysis on 8 candidate signals across 200+ tickers, surfacing lagged sector-ETF returns as the strongest predictor and shifting the trading signals team off an ARIMA baseline onto supervised learning.
- Automated a daily data preparation workflow with a Python ETL pipeline (PostgreSQL, Google Sheets API), cutting runtime from 60 minutes to 3 minutes per sync (92% reduction) and freeing 8+ analyst hours per week.
- Wrote production SQL reconciling OHLCV data across 4 source tables and added schema validation that eliminated recurring data quality errors in downstream reporting.

PROJECTS

Ragnarok: Multimodal RAG Knowledge Assistant | *Python, LangChain, ChromaDB, OpenAI, Claude, Gemini, Groq, Streamlit*

- Built a multimodal RAG pipeline ingesting PDFs, audio (Whisper transcription), and video (OpenCV frame extraction with vision-model captions) into a unified ChromaDB vector store of 10K+ chunks across 4 modalities; combined dense cosine search, BM25 sparse ranking via Reciprocal Rank Fusion, and cross-encoder re-ranking (ms-marco-MiniLM) to reduce irrelevant context passed to the generation layer by approximately 60%.
- Routed queries across 4 LLM providers (Groq, OpenAI, Gemini, Claude) behind a citation-enforcing system prompt, deployed via Streamlit with production caching, async batch embedding, and configurable retrieval parameters.

Dynamic SQL Assistant: Text-to-SQL Engine | *Python, LangChain, Groq, SQLite*

- Built a Text-to-SQL engine enabling natural language database queries with 90%+ semantic accuracy across 50+ tested patterns; designed prompt orchestration with runtime schema injection and SQL validation to guarantee syntactic correctness.

AutoML-ify: No-Code AutoML Platform | *Python, Streamlit, Scikit-learn*

- Built an end-to-end AutoML web application enabling non-technical users to train ML models without code, supporting 5+ algorithms with automated hyperparameter tuning via GridSearchCV across datasets up to 100K rows; designed an evaluation dashboard with model metrics and visualizations that cut model development time by 80% for business analyst users.

EDUCATION

University of Maryland, College Park | M.S. in Data Science *Aug 2025 – Expected May 2027*

Vellore Institute of Technology, AP | B.Tech in Computer Science and Engineering *2020 – 2024*